Answers

Advanced Algebra II Name_____

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Quadratic Word Problems

Date_____

1) A fireworks rocket is launched from a hill above a lake. The rocket will fall into the lake after exploding at its maximum height. The rocket's height above the surface of the lake is given by g(x)= $-16x^2 + 64x + 80$. What is the maximum height of the rocket?

A) 150 feet B) 80 feet
C) 120 feet D) 144 feet

$ax^2+bx+c = g(x)$

Sad face → max at vertex

Vertex at $\frac{-b}{2a} = \frac{-64}{-32} = 2$

$g(2) = -16\times2^2 + 64\times2 + 80$
$= -64 + 128 + 80$
$= 144$

Answer: D

2) A fireworks rocket is launched from a hill above a lake. The rocket will fall into the lake after exploding at its maximum height. The rocket's height above the surface of the lake is given by g(x)= $-16x^2 + 64x + 80$. How long will it take the rocket to hit the lake?

A) 5 seconds B) 8 seconds
C) 12 seconds D) 10 seconds

Solving for $-16x^2 + 64x + 80 = 0$ (When height is 0 m above ground).

Simplify ($\div(16)$) $x^2 - 4x - 5 = 0$

Factorise (or Formula) $(x-5)(x+1) = 0$

$x_1 = 5$
$x_2 = -1$ [(-1) is not possible]

Answer: A

3) A rock is thrown from the top of a tall building. The distance, in feet, between the rock and the ground x seconds after it is thrown is given by $f(x) = -16x^2 - 4x + 382$ How long after the rock is thrown is it 340 feet from the ground?

A) 2.256 seconds B) 1.50 seconds
C) 3.324 seconds D) .236 seconds

Solve for: $340 = -16x^2 - 4x + 382$

Simplify $0 = -16x^2 - 4x + 42$

$0 = -8x^2 - 2x + 21$

Formula: $x = \frac{\pm\sqrt{2^2-4(-8)21}}{-16} = \frac{\pm\sqrt{676}}{-16}$

$x = \frac{2 \pm 26}{-16}$ $x_1 = \frac{28}{-16}$ (neg, not possible)

$x_2 = \frac{-24}{-16} = \frac{3}{2} = 1.5$ s

Answer: B

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4) A rock is thrown from the top of a tall building. The distance, in feet, between the rock and the ground x seconds after it is thrown is given by $f(x) = -16x^2 - 4x + 382$ How long after the rock is thrown does it hit the ground?

A) 12.256 seconds B) 11.50 seconds
C) 4.7324 seconds D) 7.236 seconds

$0 = -16x^2 - 4x + 382$

Formula: $x = \frac{4 \pm \sqrt{2^2 - 4(-16)382}}{-32}$

$x_1 = \frac{4 + 156.4}{-32}$ [NEG!] $x_2 = \frac{4 - 156.4}{-32} = 4.8$ s

Answer C

5) In an effort to catch a criminal; a superhero is going to leap over a building and take a short cut down the ally. The function $f(x) = -16x^2 + 150x$ gives the superhero's height in feet as a function of time. The building is 425 feet high. Will the superhero make it over the building?

A) Yes, the superhero always makes it!
B) No, the superhero can only jump half the height of the building
C) No, the superhero will crash into the building at 351 feet
D) No, the superhero can only jump 150 feet into the air

Max height at vertex-time $\frac{-b}{2a} = \frac{-150}{-32} = 4.6875$ s.

Height at vertex: $-16 \times 4.6875^2 + 150 \times 4.6875 = 351.6$ m

$351 < 425$ No, he will not make it over the building. Answer C

6) The height in feet of the curved arch support for a bridge can be modeled by $f(x) = -0.0009x^2 + 1.24x + 1.65$. You are standing on a platform 2 feet above the maximum height of the arch. If you bungee from this point, and your bungee will stretch to 420 feet before retracting; are you safe to jump?

A) Yes you are totally safe!
B) No, you will hit the ground before your bungee retracts
C) Who cares! you have insurance!
D) I'm not sure, I never thought I would have to use math!

x value at vertex: $\frac{-b}{2a} = \frac{-1.24}{-0.0018} = 689$

height at vertex: $-0.0009 \times 689^2 + 1.24 \times 689 + 1.65$

The platform → (+2)

$= 430.8$ feet. > 420 feet

Yes, you are safe. Answer A

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7) You are interested in retrieving a souvenir from a shipwreck located 115 feet below the water. You do not have diving equipment, so your dive is limited by the equation $y = 0.05x^2 - 4x - 38$ Assumming you can hold your breath for the duration of the dive; will you be able to retrieve your souvenir?

A) No, you can only go 38 feet below the water level
B) No, the shipwreck is located 4 feet below your maximum depth
C) Yes, you will be able to retrieve the souvenir
D) No, you are too afraid of sharks to even attempt the dive!



Depth = $y = 0.05x^2 - 4x - 38$

Max depth at vertex. $\frac{-b}{2a} = \frac{4}{0.1} = 40$

Max depth: $0.05 \times 40^2 - 4 \times 40 - 38 = -118$ feet

Yes, you will be able to. Answer C

8) You are interested in retrieving a souvenir from a shipwreck located 115 feet below the water. You do not have diving equipment, so your dive is limited by the equation $y = 0.05x^2 - 4x - 38$ Assumming you are able to retrieve your souvenir, how long do you have to hold your breath until you resurface?

A) almost 90 seconds
B) 38 seconds
C) almost 2 minutes
D) a minute and 10 seconds

Find x when $y = 0$ — when you submerge and emerge.

$0 = 0.05x^2 - 4x - 38$

$x = \frac{4 \pm \sqrt{16 - 4 \times 0.05 \times (-38)}}{0.1} = \frac{4 \pm 4.86}{0.1}$

$x_1 = -8.58$ s

$x_2 = 88.6$ s

Total time = $x_2 - x_1 = 97$ s.

97 s in the water

89 s without air!

Answer: A

Factor each completely.

9) $6a^2 - 23a - 18$

$(3a + 2)(2a - 9)$

10) $9x^2 - 56x + 12$

$(9x - 2)(x - 6)$

Solve each equation by factoring or completing the square.

11) $3x^2 = 26x - 48$

$x^2 - \frac{26}{3} + 16 = 0$

$\left(x - \frac{13}{3}\right)^2 - \frac{25}{9} = 0$

$x - \frac{13}{3} = \pm \frac{5}{3}$

$x_1 = 6 \quad x_2 = \frac{8}{3}$

12) $7x^2 = -60x - 32$

$x^2 + \frac{60x}{7} + \frac{32}{7} = 0$

$\left(x + \frac{30}{7}\right)^2 - \frac{224}{49} = 0$

$x + \frac{30}{7} = \pm \frac{26}{7}$

$x_1 = \frac{-4}{7} \quad x_2 = \frac{-56}{7} = -8$

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13) $3r^2 - 12r = 15$

$r^2 - 4r - 5 = 0$

$(r-2)^2 - 9 = 0$

$r - 2 = \pm 3$

$r_1 = 5 \quad r_2 = 1$

14) $2n^2 - 12 = 5n$

$n^2 - \frac{5}{2}n - 6 = 0$

$(n - \frac{5}{4})^2 - \frac{121}{16} = 0$

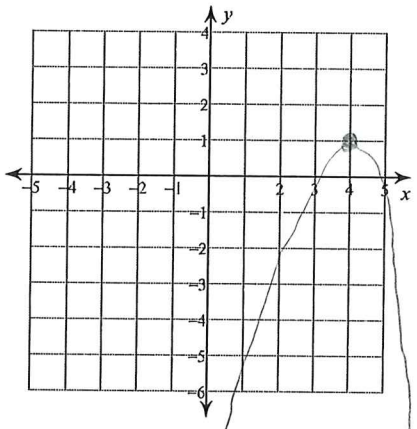
$n - \frac{5}{4} = \pm \frac{11}{4}$

$n_1 = \frac{16}{4} = 4$

$n_2 = \frac{-6}{4} = -\frac{3}{2}$

For each function, a) determine if it opens up or down, b) find the axis of symmetry, c) find the vertex, d) find the y - intercept, e) graph the function, f) determine if it has a maximum or minimum and what that value is, and g) identify the domain and range.

15) $f(x) = -2x^2 + 12x - 15$



a) ∩ Opens down

b) $\frac{-b}{2a} = \frac{-12}{-4} = 4$

c) $f(4) = -2 \times 16 + 12 \times 4 - 15$

$= -32 + 48 - 15$

$= 1$

Thus $(4, 1)$

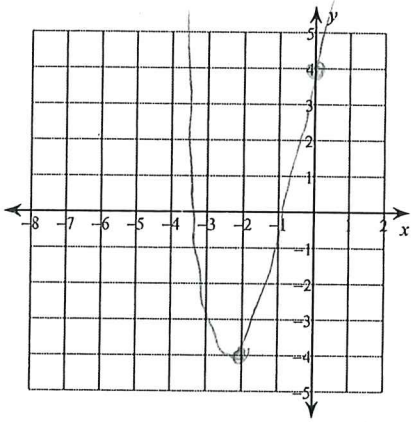
d) $y = -15$

f) max at $(4, 1)$

g) Domain $-\infty < x < \infty$

Range $-\infty < y \leq 1$

16) $f(x) = 2x^2 + 8x + 4$



a) ∪ Opens up

b) $\frac{-b}{2a} = \frac{-8}{4} = -2$

c) $f(-4) = 2 \times 4 - 16 + 4$

$= -4$

Thus $(-2, -4)$

d) $y = 4$

f) min at $(-2, -4)$

g) Domain $-\infty < x < \infty$

Range $-4 \leq y < \infty$

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